S2G7 Ch.5 Linear Equations in Two Unknowns **Short Quiz #10**

Name: ________________ Time: 12 min

Class (no.): ________ (   ) Score: 9

Date: ________________

1. Make t the subject of the formula $s = \dfrac{2t}{5t+1}$. (3 marks)

$s = \dfrac{2t}{5t+1}$

$s(5t+1) = 2t$

$5st + s = 2t$

$s = 2t - 5st$

$s = t(2-5s)$

$t = \dfrac{s}{2-5s}$

2. Solve $\begin{cases} x+2y=11 \\ 2x-3y=-6 \end{cases}$. (3 marks)

$\begin{cases} x+2y=11 & —(1) \\ 2x-3y=-6 & —(2) \end{cases}$

$2\times(1)$: $2x+4y=22$ —(3)

$(3)-(2)$ $7y=28$

$y=4$

Sub. $y=4$ into (1)

$x+2(4)=11$

$x=3$

$\therefore x=3,\ y=4$ or $\begin{cases} x=3 \\ y=4 \end{cases}$

3. There is a total of 30 \$10 coins and \$1 coins in a box. If the total value of these coins is \$120, find the numbers of \$10 and \$1 coins respectively in the box. (3 marks)

Let the no. of \$1-coins be x and that of \$10-coins be y.

$\begin{cases} x+y=30 & —(1) \\ 10y+x=120 & —(2) \end{cases}$

$(2)-(1)$: $9y=90$

$y=10$

Sub. $y=10$ into (1)

$x+10=30$

$x=20$

$\therefore$ There are 20 \$1-coins and 10 \$10-coins